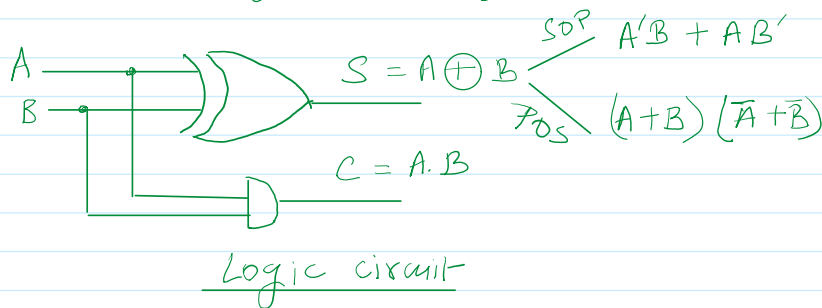


Combinational Circuits

30 July 2026 05:40

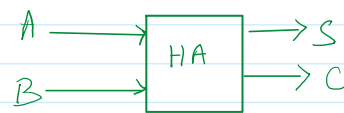
Combinational circuits have multiple inputs and multiple outputs. Examples, Half-adder, full adder, encoder, decoder, multiplexer, flip-flop, etc.

Half adder : Half adder is a combinational logic circuit which is designed by connecting one XOR gate and 1 and gate. The half adder circuit has two inputs : A and B, which add two input binary digits and generates/output carry and sum.



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

↑ XOR ↑ AND



Block diagram of HA

XOR — NAND
 — NOR
 — Basic gates (AND, OR, NOT)

H.W : Draw HA circuit using only 1. Basic gates
 2. NAND gates
 3. NOR gates.

Full adder : Full adder is the circuit which consist of 1 XOR gate, three AND gates and one OR gate. It adds 3-binary bits, and generate/output sum and carry

$$C = \overline{A} \overline{B} C + \overline{A} B C + A \overline{B} C + A B C$$

X	Y	Z	S	C
0	0	0	0	0

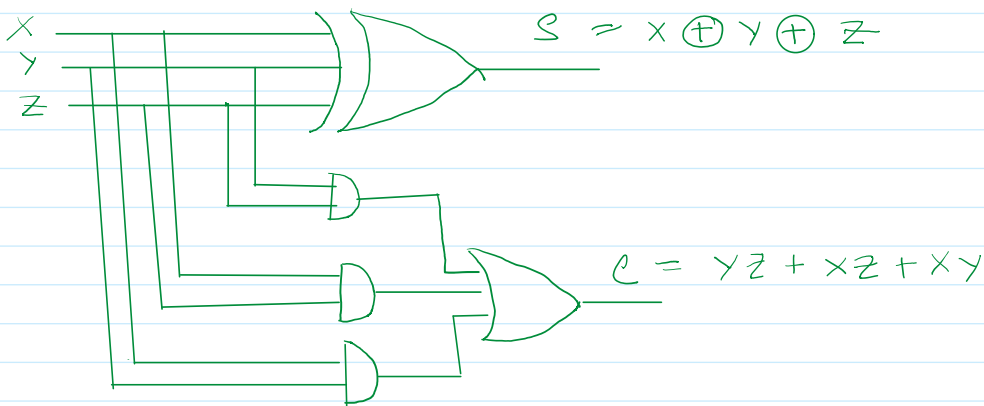
$$\begin{aligned}
 C &= \bar{X}YZ + X\bar{Y}Z + XY\bar{Z} + XYZ \\
 \text{or } C &= \bar{X}YZ + X\bar{Y}Z + XY(\bar{Z} + Z) \\
 \text{or } C &= \bar{X}YZ + X\bar{Y}Z + XY \quad [\because A + \bar{A} = 1] \\
 \text{or } C &= \bar{X}YZ + X(\bar{Y}Z + Y) \\
 \text{or } C &= \bar{X}YZ + X(Y + \bar{Y})(Y + Z) \quad [\text{distributive law}] \\
 \text{or } C &= \bar{X}YZ + XY + XZ \\
 \text{or } C &= Y(\bar{X}Z + X) + XZ \\
 \text{or } C &= X(X + \bar{X})(Y + Z) + XZ \\
 \text{or } C &= XY + YZ + XZ
 \end{aligned}$$

X	Y	Z	S	C	
0	0	0	0	0	
0	0	1	1	0	
0	1	0	1	0	
0	1	1	0	1	$\bar{X}YZ$
1	0	0	1	0	
1	0	1	0	1	$X\bar{Y}Z$
1	1	0	0	1	$XY\bar{Z}$
1	1	1	1	1	XYZ

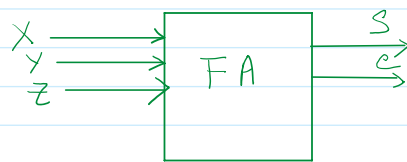
↑
XOR

OR

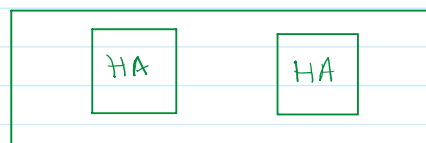
$$\begin{aligned}
 C &= \bar{X}YZ + X\bar{Y}Z + XY\bar{Z} + XYZ + X\bar{Y}Z + XYZ \\
 &= YZ(\bar{X} + X) + XZ(\bar{Y} + Y) + XY(\bar{Z} + Z) \\
 &= YZ + XZ + XY
 \end{aligned}$$



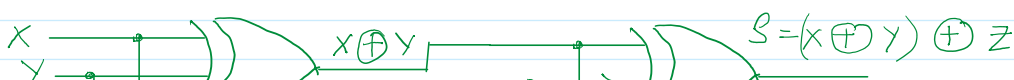
Logic ckt. diagram of fnu adder

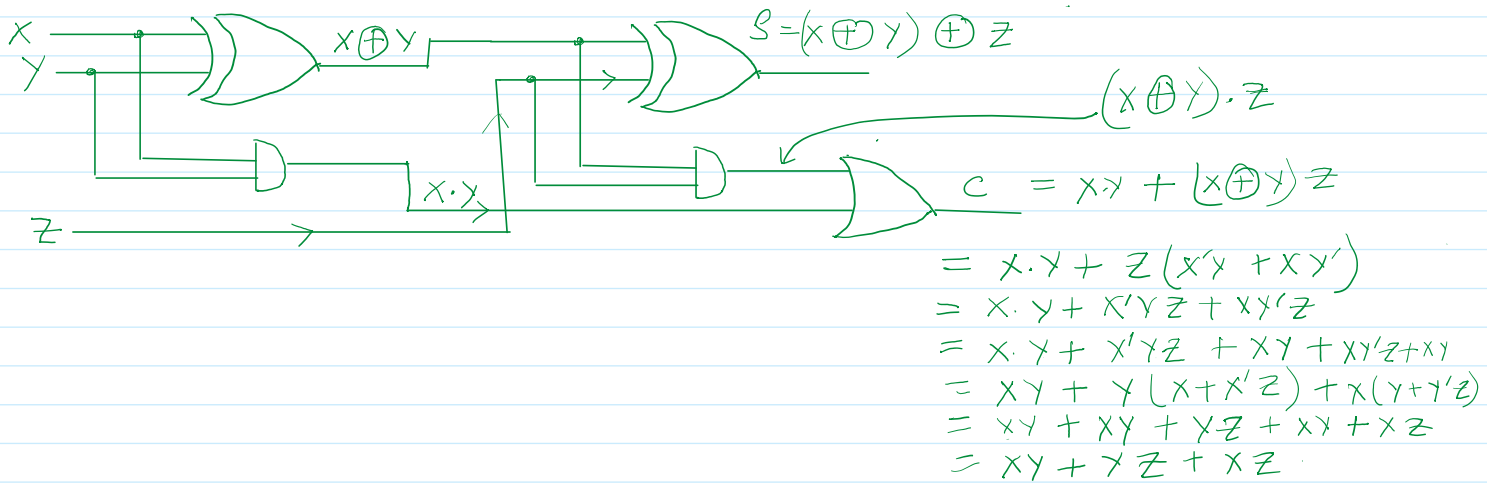


Block diagram of full-adder



Full adder





H.W Draw full adder ckt using only
 1. Basic gates
 2. NAND gates
 3. NOR gates.